Homework 2 - Due at 11:59 pm, Nov 14, 2026

I suggest using Python for this, but one can use MATLAB too.

Problem 1: 20 points

In this problem, you need to classify a breast cancer dataset: M=Malignant vs B=Benign. Use the dataset "Breast Cancer Wisconsin (Diagnostic)", which is available here in the UCI website. The dataset is also uploaded in Kaggle. You have 10 features and 569 samples. Use 75% data for training and 25% for testing. Report the confusion matrices and the corresponding sensitivity, specificity and accuracy for SVM and kNN. You need to check with different kernels (for SVM) and k (for kNN) to achieve the best result.

Problem 2: 30 points

In this problem, you need to classify an EEG dataset for seizure detection (classification). The time-series dataset can be obtained here. Download Sets A, D and E, where set A corresponds to healthy people, set D corresponds to seizure patients' seizure free (interictal) intervals, and set E corresponds to seizure patients' seizure-attack moments. There are 100 samples for each of the sets. (1) Plot one sample from each of the sets. (2) First you should find the variance for each of the time-series samples. You will use this as the discriminating feature for the classification. (3) Now, classify A vs D, A vs E and D vs E with SVM and k-NN. (4) Finally report the confusion matrices and the corresponding sensitivity, specificity and accuracy values.

Problem 3: 30 points

In this problem, you need to classify a handwritten digit dataset: MNIST, which is available here. The dataset is also uploaded in Kaggle. You have to classify images into 10 classes: train with $60k$ images, and test with $10k$ images. Show one image from each of the digit-sets. Use fully connected neural network: try with different number of hidden layers, neurons, different activation functions to obtain the best result. Plot the training loss with epochs, and also report the training time and test accuracy.

Problem 4: 30 points

In this problem, you need to classify the Fashion-MNIST dataset. The dataset is also available in Kaggle. You have to classify images into 10 classes: train with 60k images, and test with 10k images. Show one image from each of the digit-sets. Use fully connected neural network: try with different number of hidden layers, neurons, different activation functions to obtain the best result. Plot the training loss with epochs, and also report the training time and test accuracy.